



Spring Semester course

CH31010: Mass Transfer II

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L3: Diffusion

- Mass transfer may occur in a gas mixture, a liquid solution or solid.
- Mass transfer occurs whenever there is a **gradient in the concentration** of a species.
- The basic mechanisms are the same whether the phase is a gas, liquid, or solid.

Diffusion phenomena

Mass diffusion – Fickian type (molecular diffusion)

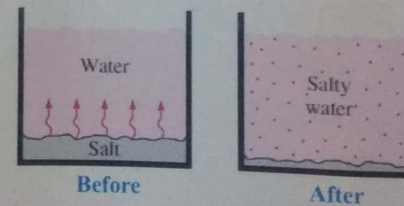
Brownian diffusion

Turbulent / eddy diffusion

Diffusion in electrolytic solution

Knudsen diffusion

- Whenever there is concentration difference in a medium, nature tends to equalize things by forcing a flow from the high to the low concentration region.



- The molecular transport process of mass is characterized by the general equation:

$$\text{Rate of transfer process} = \frac{\text{driving force}}{\text{resistance}}$$

$$N = \frac{D_A}{S} \frac{dc_A}{dz}$$

$$J_A = -D_{AB} \frac{dc_A}{dz}$$

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Molecular Diffusion Equation

- Fick's Law

$$J_{AZ}^* = -D_{AB} \frac{dc_A}{dz}$$

$$J_{AZ}^* = -\nabla (D_{AB} c_A)$$

J_A^* is the molar flux of component A in the z direction in kg mol A/s.m².

D_{AB} is the molecular diffusivity of the molecule A in B in m²/s

c_A is the concentration of A in kg mol/m³.

z is the distance of diffusion in m

Fick's Law of Diffusion

- Molecular diffusion or molecular transport can be defined as the transfer or movement of individual molecules through a fluid by mean of the random, individual movements of the molecules.

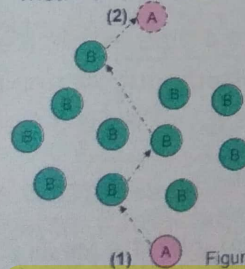


Figure 3: Schematic diagram of molecular diffusion process

- If there are greater number of A molecules near point (1) than at (2), then since molecules diffuse randomly in both direction, more A molecules will diffuse from (1) to (2) than from (2) to (1).
- The net diffusion of A is from high to low concentration regions.

Molecular diffusion

The diffusion of molecules when the whole bulk fluid is not moving but stationary. Diffusion of molecules is due to a concentration gradient.

The general Fick's Law Equation for **binary mixture of A and B**

$$J_{AZ}^* = -cD_{AB} \frac{dx_A}{dz}$$

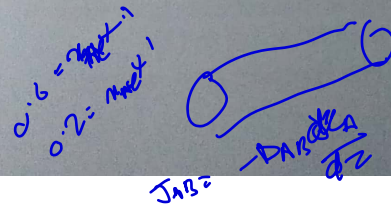
c = total concentration of A and B [kgmol (A + B)/m³]

x_A = mole fraction of A in the mixture of A and B

Example:

A mixture of He and N₂ gas is contained in a pipe at 298 K and 1 atm total pressure which is constant throughout. At one end of the pipe at point 1 the partial pressure p_{A1} of He is 0.6 atm and at the other end 0.2 m $p_{A2} = 0.2$ atm.

Calculate the flux of He at steady state if D_{AB} of the He-N₂ mixture is 0.687×10^{-4} m²/s.



$$p_A = n_A R T$$

$$\frac{p_A}{T} = \frac{n_A R}{V}$$

Solution:

- Since a total pressure P is constant, the c is constant, where c is as follows for a gas according to the perfect gas law:

$$PV = nRT$$

$$\frac{n}{V} = \frac{P}{RT} = c$$

- Where n is kg mol A plus B , V is volume in m^3 , T is temperature in K, R is $8314.3 \text{ m}^3 \cdot \text{Pa} / \text{kg mol} \cdot \text{K}$ or R is $82.057 \times 10^{-3} \text{ cm}^3 \cdot \text{atm} / \text{g. mol. K}$, and c is kg mol A plus B / m^3 .
- For steady state the flux J_{Az}^* in Eq.(6.1-3) is constant. Also D_{AB} for gas is constant. Rearranging Eq. (6.1-3) and integrating.

$$J_{Az}^* \int_{z_1}^{z_2} dz = -D_{AB} \int_{c_{A1}}^{c_{A2}} dc_A$$

$$J_{Az}^* = \frac{D_{AB}(c_{A1} - c_{A2})}{z_2 - z_1} \quad (6.1-11)$$

- Also, from the perfect gas law, $p_A V = n_A RT$, and

$$c_{A1} = \frac{p_{A1}}{RT} = \frac{n_A}{V}$$

- Substituting Eq. (6.1-12) into (6.1-11),

$$J_{Az}^* = \frac{D_{AB}(p_{A1} - p_{A2})}{RT(z_2 - z_1)} \quad (6.1-13)$$

- This is the final equation to use, which is in a form eqsily used for gases. Partial pressures are $p_{A1} = 0.6 \text{ atm} = 0.6 \times 1.01325 \times 10^5 = 6.04 \times 10^4 \text{ Pa}$ and $p_{A2} = 0.2 \text{ atm} = 0.2 \times 1.01325 \times 10^5 = 2.027 \times 10^4 \text{ Pa}$. Then, using SI units,

$$J_{Az}^* = \frac{(0.687 \times 10^{-4})(6.08 \times 10^4 - 2.027 \times 10^4)}{8314(298)(0.20 - 0)}$$

$$= 5.63 \times 10^{-6} \text{ kg mol A/s} \cdot \text{m}^2$$

Mass transfer driving force

At constant T & P, species i experiences a force (per mole), acting in the direction of z

$$F_i = - \left(\frac{\partial \mu_i}{\partial z} \right)_{T,P}$$

The chemical potential for component i in ideal phases:

For gas: $\mu_i = \mu_i^0 + RT \ln P_i = \mu_i^0 + RT \ln y_i P \rightarrow d\mu_i = RT \frac{dy_i}{y_i}$

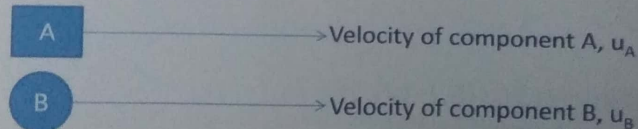
For liquid: $\mu_i = \mu_i^* + RT \ln x_i \rightarrow d\mu_i = RT \frac{dx_i}{x_i}$

For non-ideal phases: use fugacity (gas) or activity (liquid) coefficients for correction.

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Frictional resistance

Consider two components A and B:



Frictional resistance exerted by A to one mole of B, F_B , is found to be proportional to difference in the velocities:

$$F_B = -RT \frac{x_A}{D_{BA}} (u_B - u_A)$$

We will see that the constant D is in fact diffusion coefficient.

Similarly, for the force exerted by B to 1 mole of A:

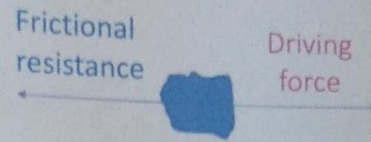
$$F_A = -RT \frac{x_B}{D_{AB}} (u_A - u_B)$$

Generalising to multi-components: $F_i = -RT \sum_{j=1}^{n-1} \frac{x_j}{D_{ij}} (u_i - u_j)$

In a unit volume of the mixture, the forces will be:

On A: $x_A C_T F_A = -RT \frac{x_B x_A C_T}{D_{AB}} (u_A - u_B)$ On B: $x_B C_T F_B = -RT \frac{x_A x_B C_T}{D_{BA}} (u_B - u_A)$

Complete picture



$$-\left(\frac{\partial \mu_i}{\partial z}\right)_{T,P} - RT \sum_{j=1}^{n-1} \frac{x_j}{D_{ij}} (u_i - u_j) = 0$$

Use fluxes and concentrations instead of velocities and mol fractions

$$-\left(\frac{\partial \mu_i}{\partial z}\right)_{T,P} = -RT \frac{1}{x_i} \frac{\partial x_i}{\partial z} = -RT \frac{1}{c_i} \frac{\partial c_i}{\partial z}$$

$$x_i = c_i / c_T, \quad c_T = c_A + c_B$$

$$N_i = c_i u_i$$

$$-\frac{\partial c_i}{\partial z} = \sum_{j=1}^{n-1} \frac{1}{c_T D_{ij}} (c_j N_i - c_i N_j)$$

Diffusion in a binary mixture

$$-\frac{\partial c_i}{\partial z} = \sum_{j=1}^{n-1} \frac{1}{c_T D_{ij}} (c_j N_i - c_i N_j)$$



$$-\frac{\partial c_B}{\partial z} = \frac{1}{c_T D_{AB}} (c_A N_B - c_B N_A)$$

$$-\frac{\partial c_A}{\partial z} = \frac{1}{c_T D_{AB}} (c_B N_A - c_A N_B)$$

Note that the interaction (frictional) term involves the knowledge of *relative velocities*.

We need therefore to specify another *single piece of information*, which will fix the velocity or flux relative to a fixed reference frame.

Specification 1: Equimolar counter-diffusion

$$-\frac{\partial C_A}{\partial z} = \frac{1}{C_T D_{AB}} (C_B N_A - C_A N_B) \quad \text{and} \quad -\frac{\partial C_B}{\partial z} = \frac{1}{C_T D_{BA}} (C_A N_B - C_B N_A)$$

The fluxes of A & B are equal and opposite (e.g. distillation)

$$N_A = -N_B$$

Therefore, $N_A = -D_{AB} \frac{\partial C_A}{\partial z}$ and $N_B = -D_{BA} \frac{\partial C_B}{\partial z}$

This looks like the Fick's law, but here we do not have the condition of being dilute or stagnant!

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Specification 2: Stagnant component (Stefan's condition)

$$-\frac{\partial C_A}{\partial z} = \frac{1}{C_T D_{AB}} (C_B N_A - C_A N_B) \quad \text{and} \quad -\frac{\partial C_B}{\partial z} = \frac{1}{C_T D_{BA}} (C_A N_B - C_B N_A)$$

The fluxes one component, say B is zero (e.g. membrane separation)

$$N_B = 0$$

$$\frac{\partial C_A}{\partial z} = -\frac{C_B}{C_T D_{AB}} N_A \quad \text{and} \quad \frac{\partial C_B}{\partial z} = \frac{C_B}{C_T D_{AB}} N_A$$

Stefan's law

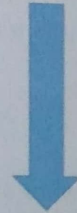
$$N_A = -\frac{C_T}{C_T - C_A} D_{AB} \frac{\partial C_A}{\partial z}$$

Note that even $N_B = 0$, the spatial gradient of B $\left(\frac{\partial C_B}{\partial z}\right)$ still exists, in the positive direction of the diffusion of A

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Specification 3: Stagnant & dilute component (Fick's law)

Stefan's law: $N_A = -\frac{c_T}{c_T - c_A} D_{AB} \frac{\partial c_A}{\partial z}$ considering the flux of one component to be zero, $N_B = 0$



Considering A to be dilute, $c_A \ll c_T$

$$N_A = -D_{AB} \frac{\partial c_A}{\partial z} \text{ (Fick's law)}$$

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Mass transport equation

$$\frac{\partial C}{\partial t} + \mathbf{v} \cdot \nabla C = \nabla \cdot (D \nabla C) + R$$

where $\mathbf{v}$ is the fluid velocity field.

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Understanding the diffusion coefficient

A typical diffusion coefficient for a molecule in the gas phase is in the range of 10^{-6} to $10^{-5} \text{ m}^2/\text{s}$. By contrast, diffusion for molecules dissolved in liquids is far slower. In an aqueous (water) solution, typical diffusion coefficients are in the range of 10^{-10} to $10^{-9} \text{ m}^2/\text{s}$. As a result, diffusion in liquids is very slow over everyday length scales and is almost always dominated by convection.

Solids: 10^{-12} – $10^{-14} \text{ m}^2/\text{s}$

Formally, the diffusion coefficient can be understood as parameterizing the area of a spherical surface, defined as the surface of root-mean-square displacement of material diffusing away from an infinitesimal point where a mass is initially concentrated. Since the statistics of diffusion cause this area to grow linearly in time, the diffusion coefficient is a quantity described by area per time.

Dependencies

Considering the values for the mean free path and average velocity for molecules in an ideal gas from the Maxwell-Boltzmann distribution, it follows that the diffusion coefficient obeys the following relation to temperature and pressure:

$$D \propto \frac{T^3}{P}$$

That is, diffusion is faster in both hotter and more rarefied gases.

For particles or large molecules in a viscous fluid (usually a liquid solution), the Stokes-Einstein equation can be applied:

$$D = \frac{kT}{6\pi\mu r}$$

Here, k is the Boltzmann constant, μ is the solvent viscosity, and r is the radius of the diffusing particle. This equation is derived on the assumption that the particles obey Stokes' law for drag, such that the drag exerted on diffusing molecules, by the solvent molecules.

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